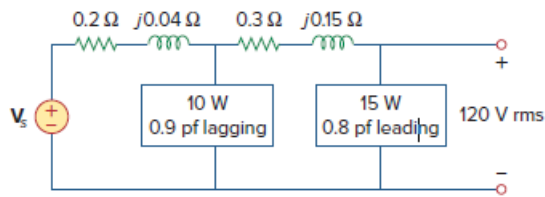


Problem

For the circuit in Fig. 11.81, find V_s .

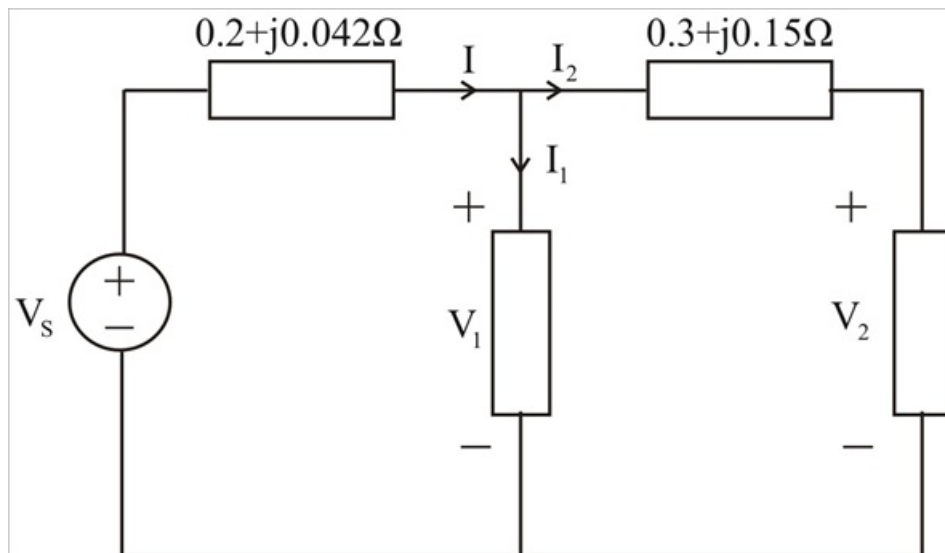
Figure 11.81:



Step-by-step solution

Step 1 of 5

Consider the circuit below.



Step 2 of 5

$$S_2 = 15 - j \left(\frac{15}{0.8} \right) \sin(\cos^{-1}(0.8))$$

$$S_2 = 15 - j11.25$$

But, $S_2 = V_2 I_2^*$

$$I_2^* = \frac{S_2}{V_2}$$

$$I_2^* = \frac{15 - j11.25}{120}$$

$$I_2^* = 0.125 - j0.09375 \text{ A}$$

$$I_2 = 0.125 + j0.09375$$

Step 3 of 5

$$V_1 = V_2 + I_2 (0.3 + j0.15)$$

$$V_1 = 120 + (0.125 + j0.09375)(0.3 + j0.15)$$

$$V_1 = 120.02 + j0.0469$$

$$S_1 = 10 + j \left(\frac{10}{0.9} \right) \sin(\cos^{-1}(0.9))$$

$$S_1 = 10 + j4.843$$

Step 4 of 5

But,

$$S_1 = V_1 I_1^*$$

$$I_1^* = \frac{S_1}{V_1}$$

$$I_1^* = \frac{10 + j4.843}{120.02 + j0.0469}$$

$$I_1^* = 0.0837 + j0.0405$$

$$I_1^* = 0.093 \angle 25.82^\circ$$

$$I_1 = 0.0837 - j0.0405$$

$$I_1 = 0.093 \angle -25.82^\circ$$

Step 5 of 5

$$I = I_1 + I_2$$

$$I = 0.2087 + j0.053$$

$$V_s = V_1 + I(0.2 + j0.04)$$

$$V_s = (120.02 + j0.0469) + (0.2087 + j0.053)(0.2 + j0.04)$$

$$V_s = 120.06 + j0.0658$$

$$\boxed{V_s = 120.06 \angle 0.03^\circ \text{ V}}$$